SWARAJ PRAVIN KUBDE

Embedded Software Engineer | Autonomous Systems | Computer Vision

swarajwork.06@gmail.com | +91 9146867271 | LinkedIn

SUMMARY

Electronics and Telecommunication Engineering student with hands-on experience in embedded software, autonomous drone systems, computer vision, and Linux-based embedded platforms. Experienced in Python, Embedded C, OpenCV, Raspberry Pi, sensor integration, and real-time image processing through industry internship and personal R&D projects. Interested in embedded software development for mobile, IoT, robotics, and edge computing systems.

PROFESSIONAL EXPERIENCE

Project Intern — Cerebrospark Innovations 01/2026 – Present | Pune

- Developed software modules for autonomous drone research involving GPS-denied navigation on Linux-based embedded platforms.
- Contributed to computer vision pipelines using Python and OpenCV for navigation, object tracking, and perception.
- Worked with Raspberry Pi Zero 2 W and Raspberry Pi 5 for sensor interfacing, camera integration, and embedded software deployment.
- Participated in development and testing of an interceptor drone platform involving real-time embedded processing. *(Keep details generic because of NDA.)*
- Assisted in mobile-to-drone communication architecture for telemetry exchange and autonomous mission control.
- Performed debugging, testing, and validation of embedded software under real-world operating conditions.
- Designed an Android-to-drone communication framework enabling telemetry exchange and remote monitoring over MAVLink-based architecture.

PROJECTS

Autonomous Quadcopter Research Platform

- Designed and assembled a custom 500 mm quadcopter using ArduPilot flight controller.
- Integrated GPS, IMU, ESCs, onboard camera, telemetry, and power distribution system.
- Developed the platform for autonomous navigation, computer vision, and embedded AI experiments.
- Performed flight testing, sensor calibration, PID tuning, and system validation.
- Designed software architecture for Android-based Ground Control Station communication.

Self-Balancing Robot Using Arduino

- Built a two-wheel self-balancing robot using Arduino UNO and MPU6050 gyro sensor.
- Implemented PID control using MPU6050 for real-time stability
- Tuned control parameters to minimize oscillations

Android Drone Ground Station

- Developed an Android application in Kotlin for real-time drone telemetry visualization.
- Implemented MAVLink communication over USB and WebSocket.
- Designed live dashboard showing battery, GPS, flight mode, altitude, and mission status.
- Implemented Air Unit and Ground Station architecture for autonomous drone communication.

SKILLS

Programming — Python, Embedded C, JavaScript

Computer Vision & Robotics — OpenCV, Digital Image Processing, Autonomous Systems, PID Control

Embedded Systems — Linux Raspberry Pi Arduino ESP32 ESP8266 UART I2C SPI MAVLink

Tools & Software — Git GitHub Mission Planner QGroundControl MATLAB Fusion 360

EDUCATION

B.E. Electronics & Telecommunication Engineering
Trinity Academy of Engineering College, Pune
2023 – 2027
CGPA: FE 9.27 | SE 9.20

LEADERSHIP & ACTIVITIES

- **GDGOC-TAE Management Team Member** — Organized technical workshops and community events